

UCE306: ARCHITECTURE DRAWING AND BUILDING CONSTRUCTION

L	T	P	Cr
2	0	2	3.0

Course Objective: To expose students to the concepts of architectural drawings and building construction.

Introduction to Architecture Drawing: Proportion, orientation, building bye laws, site plan, building layout, Architectural and structural working drawings, Modular co-ordination and drawing on modules,

Building structures: Various building components in load bearing and framed structures.

Building information modelling (BIM): Introduction to BIM and basics of modelling.

Foundations: *Shallow foundations:* Continuous, spread, arch, combined, cantilevered, Raft, Grillage, *Deep foundations:* Piles & wells, Footings in block cotton soil, Basement & Retaining walls

Masonry: Stone, Brick & Concrete Block masonry, Bonds and junctions, Walling, Mud wall, Sun-dried bricks, burnt bricks, stones walling, load bearing & non load bearing brick masonry, panel wall and reinforced masonry for multi-storied construction.

Special treatments: Damp Proof Course, points of its requirement in buildings, D.P.C. at Plinth level, in basement and roof tops etc., waterproofing treatments, anti-termite treatment, heat and acoustic insulation of buildings

Lintels & Arches: Location and construction details in wood, brick, stone and R.C.C.

Stairs & Stair cases: Suitability of location, stairs in multistoried buildings, Residential and public buildings, Fire escape, Stairs in timber, stone, brick, RCC and Metal Drawings in Plan elevation and sections. Hand rail & railings, description and sketches of lifts escalators.

Doors & Windows: Details, Location in buildings, sizes, Door-window schedule & construction for wooden & metal, Battened braced, framed, flush and paneled, sliding, folding telescopic, with louvers, collapsible, Revolving doors, Aluminum doors, etc. Windows in timber, aluminium frame & metal casement, double hung, Dormer, Corner, Fanlight, skylight, clear storey etc., and low-cost ideas.

Roofing and Flooring: Types of flooring, Flat roofs: Waffle floor, wooden flooring, channels, cored units etc, inclined roofs.

Miscellaneous: Formwork, scaffolding, underpinning.

Precast and Prefabricated Elements: Pre-engineered buildings (PEB), Prefabricated Prefinished Volumetric Construction (PPVC), Prefabricated components, Assembly at site, joints in prefabricated construction, low-cost housing & hollow blocks.

Laboratory Work

Modelling of building layout and building components in BIM based software

- A two-storey building plan with three/four rooms: Plan, Elevations & Section (Modular), Site Plan (Application of Bye laws), Foundation for walls – Construction details, Proposed doors and windows – Construction details, Roof & floor – details, in construction Stair case details
- Drawings of all the above building components e.g. Brick masonry bonds and junctions, DPC, Lintels and Arches, Stairs, Doors & Windows, Roof & roof coverings
- Drawing and detailing of typical building components

Course Learning Objectives (CLO)

The students will be able to:

1. Plan and draw constructional details of different building components
2. Capable of working with an architect and contractor
3. Prepare building plans and other components for a project
4. Capable of supervise building constructions

Text Books

1. Building Construction and Materials, Singh, Gurcharan, Standard Book House (2019).
2. Building Construction, Sharma, S. K., S. Chand and Company (2012).

Reference Books

1. Building Construction, Kumar, Sushil, Standard Publisher and Distributors (2020).
2. Building Construction, Punima, B. C., Laxmi Publishing House (2002).
3. A Text Book of Building Construction, Sharma and Kaul, S., Chand and Company (1987).
4. National Building Code, Volume-I and Volume-II, Bureau of Indian Standards (2016)
5. IS: 962: Code of Practice for Architectural and Building Drawings, Bureau of Indian Standards (2001)

Evaluation Scheme

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	25
2.	EST	35
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	40